

RaceDay API Endpoint Plan

PROG6212 Programming 2B - PoE Part 1

This document defines the RESTful API endpoints planned for the RaceDay Event Management System.

The API will support authentication, user profiles, event management, event categories, participant enrolments and race results.

Roles

Role	Description
Public	No authentication required
Any Authenticated User	Any logged-in RaceDay user
Organiser	Organiser role only
Participant	Participant role only

1. Authentication

HTTP Method	Route	Description	Role	Required Request Body	Expected Response
POST	/api/auth/register	Registers a new RaceDay user account.	Public	{ firstName, lastName, email, password, role, phoneNumber }	201 Created; 400 Bad Request; 409 Conflict
POST	/api/auth/login	Authenticates a registered user and returns an authentication token.	Public	{ email, password }	200 OK; 400 Bad Request; 401 Unauthorized

2. User Profile

HTTP Method	Route	Description	Role	Required Request Body	Expected Response
GET	/api/users/me	Returns the profile of the currently authenticated user.	Any Authenticated User	None	200 OK; 401 Unauthorized

HTTP Method	Route	Description	Role	Required Request Body	Expected Response
PUT	/api/users/me	Updates the profile of the currently authenticated user.	Any Authenticated User	{ firstName, lastName, phoneNumber }	200 OK; 400 Bad Request; 401 Unauthorized

3. Events

HTTP Method	Route	Description	Role	Required Request Body	Expected Response
GET	/api/events	Returns all available RaceDay events.	Public	None	200 OK
GET	/api/events/{id}	Returns details for a specific RaceDay event.	Public	None	200 OK; 404 Not Found
POST	/api/events	Creates a new RaceDay event.	Organiser	{ eventName, description, eventDate, location, distance, eventTypeid }	201 Created; 400 Bad Request; 401 Unauthorized; 403 Forbidden
PUT	/api/events/{id}	Updates an event managed by the authenticated Organiser.	Organiser	{ eventName, description, eventDate, location, distance, eventTypeid }	200 OK; 400 Bad Request; 403 Forbidden; 404 Not Found
DELETE	/api/events/{id}	Deletes an event managed by the authenticated Organiser.	Organiser	None	204 No Content; 403 Forbidden; 404 Not Found

4. Categories

HTTP Method	Route	Description	Role	Required Request Body	Expected Response
GET	/api/events/{eventId}/categories	Returns all categories belonging to a specific event.	Public	None	200 OK; 404 Not Found
GET	/api/categories/{id}	Returns the details of a specific event category.	Public	None	200 OK; 404 Not Found
POST	/api/events/{eventId}/categories	Creates a new category for an event managed by the authenticated Organiser.	Organiser	<pre>{ categoryName, description, minimumAge, maximumAge }</pre>	201 Created; 400 Bad Request; 403 Forbidden; 404 Not Found
PUT	/api/categories/{id}	Updates a category belonging to an event managed by the authenticated Organiser.	Organiser	<pre>{ categoryName, description, minimumAge, maximumAge }</pre>	200 OK; 400 Bad Request; 403 Forbidden; 404 Not Found
DELETE	/api/categories/{id}	Deletes a category from an event managed by the authenticated Organiser.	Organiser	None	204 No Content; 403 Forbidden; 404 Not Found

5. Enrolments

HTTP Method	Route	Description	Role	Required Request Body	Expected Response
POST	/api/enrolments	Enrols the authenticated Participant in an event and records the selected category.	Participant	<pre>{ eventId, categoryId }</pre>	201 Created; 400 Bad Request; 401 Unauthorized; 409 Conflict

HTTP Method	Route	Description	Role	Required Request Body	Expected Response
GET	/api/enrolments/me	Returns all event enrolments belonging to the authenticated Participant.	Participant	None	200 OK; 401 Unauthorized
GET	/api/events/{eventId}/enrolments	Returns all enrolments for an event managed by the authenticated Organiser.	Organiser	None	200 OK; 403 Forbidden; 404 Not Found
GET	/api/enrolments/{id}	Returns a specific enrolment for the authenticated user or an event managed by the Organiser.	Any Authenticated User	None	200 OK; 403 Forbidden; 404 Not Found
PUT	/api/enrolments/{id}/status	Updates the status of an enrolment when permitted by the system.	Organiser	<code>{ status }</code>	200 OK; 400 Bad Request; 403 Forbidden; 404 Not Found

6. Results

HTTP Method	Route	Description	Role	Required Request Body	Expected Response
POST	/api/enrolments/{enrolmentId}/result	Records a participant's finish time and finishing position for an enrolment.	Organiser	<code>{ finishTime, finishingPosition }</code>	201 Created; 400 Bad Request; 403 Forbidden; 404 Not Found
GET	/api/results/me	Returns the authenticated Participant's	Participant	None	200 OK; 401 Unauthorized

HTTP Method	Route	Description	Role	Required Request Body	Expected Response
		race results and performance history.			
GET	/api/enrolments/{enrolmentId}/result	Returns the result associated with a specific enrolment.	Any Authenticated User	None	200 OK; 403 Forbidden; 404 Not Found
PUT	/api/enrolments/{enrolmentId}/result	Updates the recorded result for an enrolment.	Organiser	<pre>{ finishTime, finishingPosition }</pre>	200 OK; 400 Bad Request; 403 Forbidden; 404 Not Found
DELETE	/api/enrolments/{enrolmentId}/result	Removes a result that was recorded incorrectly.	Organiser	None	204 No Content; 403 Forbidden; 404 Not Found

7. API Design Rules

Authentication and Authorisation

Protected endpoints require an authenticated RaceDay user.

Organiser-only endpoints must verify that the authenticated user has the ORGANISER role.

Participant-only endpoints must verify that the authenticated user has the PARTICIPANT role.

Data Validation

The API should validate required fields before database operations.

Event and Category identifiers must reference existing records.

A Participant must not be allowed to enrol in a Category that belongs to a different Event.

A Participant must not be allowed to enrol in the same Event more than once.

A Result must contain a valid finish time and a positive finishing position.

Response Codes

The API uses HTTP status codes to communicate the outcome of each request. Common responses include:

- 200 OK
- 201 Created
- 204 No Content
- 400 Bad Request
- 401 Unauthorized
- 403 Forbidden
- 404 Not Found
- 409 Conflict